

AMC 10 COMPREHENSIVE REFERENCE

AMC 10 Cheat Sheet

Core formulas, recurring structures, fast methods and late-question power tools — designed for no-calculator competition use.

CORE instant recall

HIGH high-yield

BONUS late-paper tool

Exam map CORE

- 25 multiple-choice problems • 75 minutes.
- Scoring: **+6 correct, +1.5 blank, 0 incorrect**.
- No calculator. Figures are not necessarily drawn to scale.
- Scope: elementary algebra, basic geometry, elementary number theory, counting/probability; **no trigonometry**.
- All questions are equal in raw value — protect early points.

Three-pass rule: secure easy points → return to medium questions → spend remaining time on the questions with a visible structure.

Universal contest moves HIGH

- **Backsolve:** test choices, often starting with C.
- **Plug in:** choose convenient values when only ratios matter.
- **Work backwards:** reverse a multi-step process.
- **Small cases:** test $n = 1, 2, 3$ to expose a pattern.
- **Bounds:** estimate before exact work.
- **Invariant:** track parity, remainders, totals or colors.

Fraction • percent • ratio

- Percent multiplier: increase $p\% \Rightarrow 1 + p/100$; decrease $p\% \Rightarrow 1 - p/100$.
- Successive percent changes **multiply**; they do not add.
- Compare a/b and c/d by comparing ad and bc .
- Part : whole = percent : 100.

$$\text{new} = \text{old} \left(1 \pm \frac{p}{100} \right)$$

Rates • averages • units

- Distance = rate $\times$ time.
- Work rates add: a job taking t hours has rate $1/t$.
- Weighted mean: weight by amount, not by number of groups.
- For equal distances at speeds u, v , average speed is the harmonic mean.

$$v_{\text{avg}} = \frac{2uv}{u + v}$$

ALGEBRA I

Expressions, equations & identities

The highest-return algebra tools for Questions 1–20.

Factoring identities CORE

$$a^2 - b^2 = (a - b)(a + b)$$

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

Before expanding, ask whether the expression factors.

Absolute value

$$|x - a| = d \iff x = a \pm d \quad (d \geq 0)$$

- $|x|$ is distance from 0.
- $|x - a|$ is distance from a .
- For inequalities, think in intervals before manipulating symbols.

Digit algebra

For digits a, b, c :

$$\overline{ab} = 10a + b, \quad \overline{abc} = 100a + 10b + c$$

Reversal: $\overline{ba} = 10b + a$. Digit sums naturally connect to divisibility by 3 and 9.

Linear equations & systems

- Elimination is fastest when coefficients nearly match.
- Substitution is fastest when one variable is already isolated.
- In symmetric systems, add/subtract equations first.
- Track restrictions before multiplying by variable denominators.

Exponents & radicals

$$a^m a^n = a^{m+n}, \quad (a^m)^n = a^{mn}, \quad a^{-n} = \frac{1}{a^n}$$

- $\sqrt{x^2} = |x|$, not automatically x .
- When rationalizing or squaring, check domain and extraneous roots.
- Huge powers often signal a repeating last-digit/remainder cycle.

Useful substitutions HIGH

- Repeated chunk? Set it equal to one variable.
- If $x + 1/x$ appears, square it rather than solving for x .

$$\left(x + \frac{1}{x}\right)^2 = x^2 + 2 + \frac{1}{x^2}$$

ALGEBRA II

Quadratics, functions & sequences

Structure beats computation: use roots, symmetry and recurrence patterns.

Quadratics CORE

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Delta = b^2 - 4ac$$

- $\Delta > 0$: 2 real roots; $= 0$: repeated root; < 0 : no real roots.
- Vertex x-coordinate: $-b/(2a)$.
- Complete the square for ranges/minima/maxima.

Vieta HIGH

If roots of $ax^2 + bx + c = 0$ are r, s :

$$r + s = -\frac{b}{a}, \quad rs = \frac{c}{a}$$

$$r^2 + s^2 = (r + s)^2 - 2rs$$

If the question asks about roots but not the roots themselves, try Vieta first.

Functions & graphs CORE

$$y = mx + b, \quad m = \frac{y_2 - y_1}{x_2 - x_1}$$

- **Domain** = allowed inputs; **range** = possible outputs.
- Zeros = x-intercepts; $f(0)$ = y-intercept.
- $f(x - h) + k$: right h , up k .
- $-f(x)$: reflect over x-axis; $f(-x)$: reflect over y-axis.
- Composition: $(f \circ g)(x) = f(g(x))$. For piecewise rules, identify the correct branch *before* substituting.

Arithmetic sequences

$$a_n = a_1 + (n - 1)d$$

$$S_n = \frac{n}{2}(a_1 + a_n)$$

For equally spaced numbers, the average is $(\text{first} + \text{last})/2$.

Geometric sequences

$$a_n = a_1 r^{n-1}$$

$$S_n = a_1 \frac{1 - r^n}{1 - r} \quad (r \neq 1)$$

Check whether a process multiplies by a constant ratio rather than adds a constant difference.

Standard sums / telescoping HIGH

$$1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$$

If consecutive denominator factors appear, try partial fractions and cancellation.

GEOMETRY I

Triangles, angles & similarity

Most AMC 10 geometry is won by recognizing a familiar structure quickly.

Angle facts CORE

- Triangle sum 180° .
- Quadrilateral sum 360° .
- Exterior angles of any polygon sum to 360° .
- Interior-angle sum of an n -gon: $(n - 2)180^\circ$.
- Parallel lines: corresponding and alternate interior angles.

Right triangles

$$a^2 + b^2 = c^2$$

Triples: 3-4-5, 5-12-13, 8-15-17, 7-24-25 and multiples.

$$45 - 45 - 90 : 1 : 1 : \sqrt{2}$$

$$30 - 60 - 90 : 1 : \sqrt{3} : 2$$

Triangle area

$$K = \frac{1}{2}bh$$

- Any side may be the base; height is perpendicular to its line.
- Same altitude $\Rightarrow$ area ratio = base ratio.
- Same base $\Rightarrow$ area ratio = height ratio.

Similarity HIGH

- AA is usually the fastest similarity test.
- Corresponding side ratios are equal.
- If side scale is k , perimeter scale is k , area scale is k^2 .
- Parallel segment inside a triangle is a similarity trigger.

Median & midpoint

- A median joins a vertex to the opposite midpoint.
- A median splits a triangle into two equal-area triangles.
- Midsegment joining two side midpoints is parallel to third side and half its length.

Angle bisector HIGH

If AD bisects $\angle A$ and $D \in BC$:

$$\frac{BD}{DC} = \frac{AB}{AC}$$

Combine this with the same-altitude area ratio immediately.

GEOMETRY II

Circles, coordinates, polygons & solids

A compact formula bank plus the geometric shortcuts AMC 10 actually rewards.

Circles CORE

$$C = 2\pi r, \quad A = \pi r^2$$

- Arc fraction = central-angle fraction of 360° .
- Sector area = same fraction of πr^2 .
- Inscribed angle = half the intercepted arc.
- Radius is perpendicular to tangent at tangency; tangents from one external point are equal.

$$PA \cdot PB = PC \cdot PD$$

BONUS Intersecting chords/secants: a product of segment lengths is often the intended shortcut.

Polygons

$$\text{diagonals} = \frac{n(n-3)}{2}$$

$$\text{regular exterior angle} = \frac{360^\circ}{n}$$

Regular polygons often reduce to isosceles triangles from the center.

Coordinate geometry

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Place geometry on axes when right angles, symmetry or midpoints dominate.

Area of common shapes

$$A_{\text{rect}} = lw, \quad A_{\text{para}} = bh$$

$$A_{\text{trap}} = \frac{1}{2}(b_1 + b_2)h$$

Composite area: add simple pieces or subtract holes.

Volumes & surface area

$$V_{\text{prism}} = Bh, \quad V_{\text{cyl}} = \pi r^2 h$$

$$V_{\text{pyramid}} = \frac{1}{3}Bh, \quad V_{\text{cone}} = \frac{1}{3}\pi r^2 h$$

$$V_{\text{sphere}} = \frac{4}{3}\pi r^3, \quad SA_{\text{sphere}} = 4\pi r^2$$

$$SA_{\text{cyl}} = 2\pi r^2 + 2\pi rh$$

Similar solids: length $k \Rightarrow$ surface area $k^2 \Rightarrow$ volume k^3 .

Geometry mode switch HIGH

- Many ratios? $\rightarrow$ similarity / area ratios.
- Messy right-angle figure? $\rightarrow$ coordinates.
- Repeated identical pieces? $\rightarrow$ symmetry.
- Unknown area surrounded by known areas? $\rightarrow$ subtract from a larger simple shape.

NUMBER THEORY

Divisibility, factors, gcd/lcm & modular thinking

Prime factorization is the master representation. Use remainders to avoid huge arithmetic.

Divisibility CORE

Divisor	Fast test
2	last digit even
3 / 9	digit sum divisible by 3 / 9
4	last 2 digits
5	ends 0 or 5
8	last 3 digits
11	alternating digit sum multiple of 11

Prime factorization

If

$$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$$

then the number of positive divisors is

$$\tau(n) = (e_1 + 1)(e_2 + 1) \cdots (e_k + 1)$$

Nuance: a number has an odd number of divisors iff it is a perfect square.

GCD / LCM CORE

$$\gcd(a, b) \operatorname{lcm}(a, b) = ab$$

- GCD: minimum prime exponents.
- LCM: maximum prime exponents.
- Euclid: $\gcd(a, b) = \gcd(b, a \bmod b)$.

Remainders / congruences HIGH

$$a \equiv b \pmod{m} \iff m \mid (a - b)$$

- Add, subtract and multiply remainders freely.
- Last digit $\rightarrow \bmod 10$.
- Parity $\rightarrow \bmod 2$.
- Digit sum $\rightarrow \bmod 3$ or 9.
- Day-of-week cycles $\rightarrow \bmod 7$.

Squares & parity

- Squares mod 4 are only 0 or 1.
- Odd squares are $1 \pmod{8}$.
- Even $\pm$ even = even; odd $\pm$ odd = even; even $\pm$ odd = odd.

Before solving a Diophantine-looking equation, test parity or a small modulus.

Integer-solution triggers HIGH

- Consecutive integers: $n, n + 1, \dots$
- Even/odd: $2k, 2k + 1$.
- For $ax + by = c$, integer solutions require $\gcd(a, b) \mid c$.
- Factorization often converts an integer equation into divisor counting.

COUNTING & PROBABILITY

Count cleanly before calculating probability

Separate order, cases, restrictions and symmetry. Most errors are counting errors, not arithmetic errors.

Product vs sum rule CORE

- Do A **and then** B → multiply.
- Do A **or** B, with no overlap → add.
- Break difficult problems into non-overlapping cases.

Permutations & combinations

$$P(n, r) = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Order matters? permutation. **Only selection?** combination.

Binomial / repeated objects HIGH

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$$

$$\binom{n}{r} = \binom{n}{n-r}$$

If n objects contain repeated groups of sizes $a, b, c, \dots$:

$$\frac{n!}{a!b!c!\dots}$$

Divide out reorderings that do not create a new arrangement.

Probability CORE

For **equally likely elementary outcomes**:

$$P(E) = \frac{\# \text{ favorable}}{\# \text{ total}}$$

$$P(E^c) = 1 - P(E)$$

- Independent events: multiply probabilities.
- Without replacement: probabilities usually change after each draw.

Complement & inclusion-exclusion HIGH

$$P(\text{at least one}) = 1 - P(\text{none})$$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Use complement for "at least one"; use inclusion-exclusion when two restrictions overlap.

Grid paths • pigeonhole • distributions HIGH

$$\text{shortest paths} = \binom{r+u}{r}$$

$$\text{some box has at least } \left\lceil \frac{N}{k} \right\rceil$$

$$x_1 + \dots + x_k = n, x_i \geq 0 \Rightarrow \binom{n+k-1}{k-1}$$

BONUS Stars & Bars is for distributing identical objects into labeled groups.

APPLICATIONS & BOUNDING

Word-problem templates, inequalities and efficient estimation

Translate the story into the right mathematical object before doing arithmetic.

Motion

$$d = rt$$

- Toward each other → relative speed $r_1 + r_2$.
- Same direction → catch-up speed $|r_1 - r_2|$.
- Always reconcile units first.

Work

$$\frac{1}{T} = \frac{1}{t_1} + \frac{1}{t_2} + \dots$$

Add **rates**, not completion times.

Mixtures / weighted average

$$\text{weighted mean} = \frac{\sum w_i x_i}{\sum w_i}$$

Think total amount of the feature = sum of contributions from each component.

Basic inequalities / bounds

- When multiplying/dividing an inequality by a negative, reverse the sign.
- $|x + y| \leq |x| + |y|$.
- If choices are far apart, prove a tight interval instead of finding an exact value.

$$xy = \left(\frac{x+y}{2}\right)^2 - \left(\frac{x-y}{2}\right)^2 \leq \left(\frac{x+y}{2}\right)^2$$

So for a fixed sum of two real numbers, the product is maximized when the two numbers are equal.

Data

$$\text{mean} = \frac{\text{sum}}{\text{count}}$$

- Median requires sorted data.
- Adding the same number to every entry adds it to the mean and median.
- Weighted averages stay between the smallest and largest component values.

Classic translation patterns

- Ages: differences stay constant.
- Digits: $10a + b$.
- Percent: use multipliers.
- Population/growth: repeated multiplication.
- Painted cubes: classify corner / edge / face / interior cubes.
- Clock/calendar: convert to modular cycles.

ADVANCED AMC 10 CONCEPTS

Q18–25 Power Tools

These are not random extra formulas. Each is a representation change that can collapse a hard problem.

1. Area-ratio reasoning

If two triangles share an altitude:

$$\frac{[T_1]}{[T_2]} = \frac{b_1}{b_2}$$

If they share a base:

$$\frac{[T_1]}{[T_2]} = \frac{h_1}{h_2}$$

A point dividing a side in ratio $m : n$ divides the two adjacent triangle areas in ratio $m : n$.

2. Similarity chaining

- Parallel lines $\rightarrow$ AA similarity.
- Write one consistent correspondence before using ratios.
- Side scale $k \rightarrow$ area scale k^2 .

If a length chase becomes long, ask whether the problem really wants an area or ratio.

3. SFFT / hidden factorization

For integer equations containing xy, x, y , complete a rectangle:

$$xy + ax + by = c \Rightarrow (x + b)(y + a) = c + ab$$

Then integer solutions become factor pairs.

4. Modular cycles

- Compute a few powers until the remainders repeat.
- Use the cycle length to reduce a huge exponent.
- Choose mod 2, 3, 4, 5, 8, 9, 10 according to the structure.

5. Invariants & extremal thinking

- **Invariant:** find a quantity every legal move preserves.
- **Extremal:** inspect the largest/smallest/leftmost object and exploit what must be true.
- **Work backwards:** especially effective for reversible processes.

6. Counting mode switches

- "At least one" $\rightarrow$ complement.
- Overlapping restrictions $\rightarrow$ cases / inclusion-exclusion (2-set form).
- Guaranteed repetition $\rightarrow$ pigeonhole.
- Short grid path $\rightarrow$ choose positions of identical moves.

FINAL REVIEW

Formula sprint + common traps

The page to scan immediately before a timed set.

Formula sprint

- $a^2 - b^2 = (a - b)(a + b)$
- $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- Vieta: $r + s = -b/a$, $rs = c/a$
- $a^2 + b^2 = c^2$
- $K = bh/2$
- Circle: $C = 2\pi r$, $A = \pi r^2$
- Polygon interior sum: $(n - 2)180^\circ$
- $\tau(n) = \prod (e_i + 1)$
- $\gcd \cdot \text{lcm} = ab$
- $\binom{n}{r} = n! / [r!(n - r)!]$

Common traps HIGH

- Side ratio k gives area ratio k^2 , not k .
- Average speed is not usually the arithmetic mean.
- +20% then -20% is not zero net change.
- $\sqrt{x^2} = |x|$.
- Squaring can create extraneous roots.
- Without replacement is usually dependent.
- Do not trust a geometry diagram to scale.

Recognition triggers

- Parallel lines $\rightarrow$ similarity.
- Point splits a triangle side $\rightarrow$ area ratio.
- Huge exponent $\rightarrow$ modular cycle.
- $xy + x + y$ -type expression $\rightarrow$ SFFT.
- "At least one" $\rightarrow$ complement.
- Repeated moves $\rightarrow$ invariant.
- Messy geometry + right angles $\rightarrow$ coordinates.

Final 20-second checklist

- ☐ Did I answer the quantity actually asked?
- ☐ Are my units consistent?
- ☐ Did I respect integer/positive/distinct restrictions?
- ☐ Is my magnitude plausible?
- ☐ Can I eliminate choices by parity or bounds?
- ☐ If I squared or cleared denominators, did I check the answer?